CYBERNAUT
ENDEAVOUR TO EXPLORE

TECH TRIO INTERNSHIP

WEB SCRAPING
MINI PROJECT – 1

IMDB MOVIE RATING SCRAPPER

TEAM NUMBER 6

Submitted by

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1.ABSTRACT

➢ The IMDb Top 250 Movie Rating Website is a data-driven application that uses IMDb's official datasets to collect and process movie titles, release years, and ratings. Using Python and Pandas, the system filters and ranks movies to generate a Top 250 list, which is stored in CSV format. A Flask-based website allows users to view, search, and sort movies by title, year, rank, and rating. The project demonstrates data processing and web development using Python, Flask, HTML, CSS, and JavaScript.

2.PROJECT DESCRIPTION

- Python-based IMDb Top 250 movie data application.
- Uses IMDb official datasets for movie details and ratings.
- Filters and ranks movies based on ratings and votes.

- Stores processed data in CSV format.
- Flask website displays the Top 250 movies.
- Provides search and sorting options by rank, title, year, and rating.
- Developed using Python, Pandas, Flask, HTML, CSS, and JavaScript.

3.OBJECTIVES

- To collect movie details and ratings from IMDb datasets.
- To filter and rank movies based on ratings and votes.
- To generate an accurate Top 250 movie list.
- To store processed movie data in CSV format.
- To develop a user-friendly Flask web application.
- To provide search and sorting features for easy movie analysis.
- To demonstrate data processing and web development using Python.

4.TECHNOLOGIES USED

- Programming Language: Python
- Data Processing: Pandas
- Web Framework: Flask
- Frontend: HTML, CSS, JavaScript
- Data Source: IMDb Official Datasets
- Data Storage: CSV
- Development Tool: Visual Studio Code / Python IDLE

5.FEATURES

- Collects movie details from IMDb official datasets.
- Generates the Top 250 movie ranking list.
- Displays movie title, release year, rank, and IMDb rating.
- Stores processed data in CSV format.
- Provides movie search functionality.

- Supports sorting by rank, title, year, and rating.

6.APPLICATIONS

- Helps users discover highly rated movies.
- Useful for movie comparison and selection.
- Supports movie data analysis and visualization.
- Can be used by movie review and recommendation websites.
- Helps researchers analyse movie ratings and trends.
- Useful for entertainment platforms and film enthusiasts.
- Can be extended to build personalized movie recommendation systems.

7.IMPLEMENTATION

- Collect IMDb datasets.
- Process data using Python and Pandas.

- Filter and rank the Top 250 movies.
- Save data in CSV format.
- Display movies using Flask.
- Add search and sorting features.

8.OUTCOMES

- Generated a ranked list of the IMDb Top 250 movies.
- Created a structured CSV dataset for analysis.
- Developed a user-friendly movie listing website.
- Enabled movie search and sorting functionality.
- Improved accessibility to movie ratings and details.
- Demonstrated practical use of Python, Pandas, and Flask

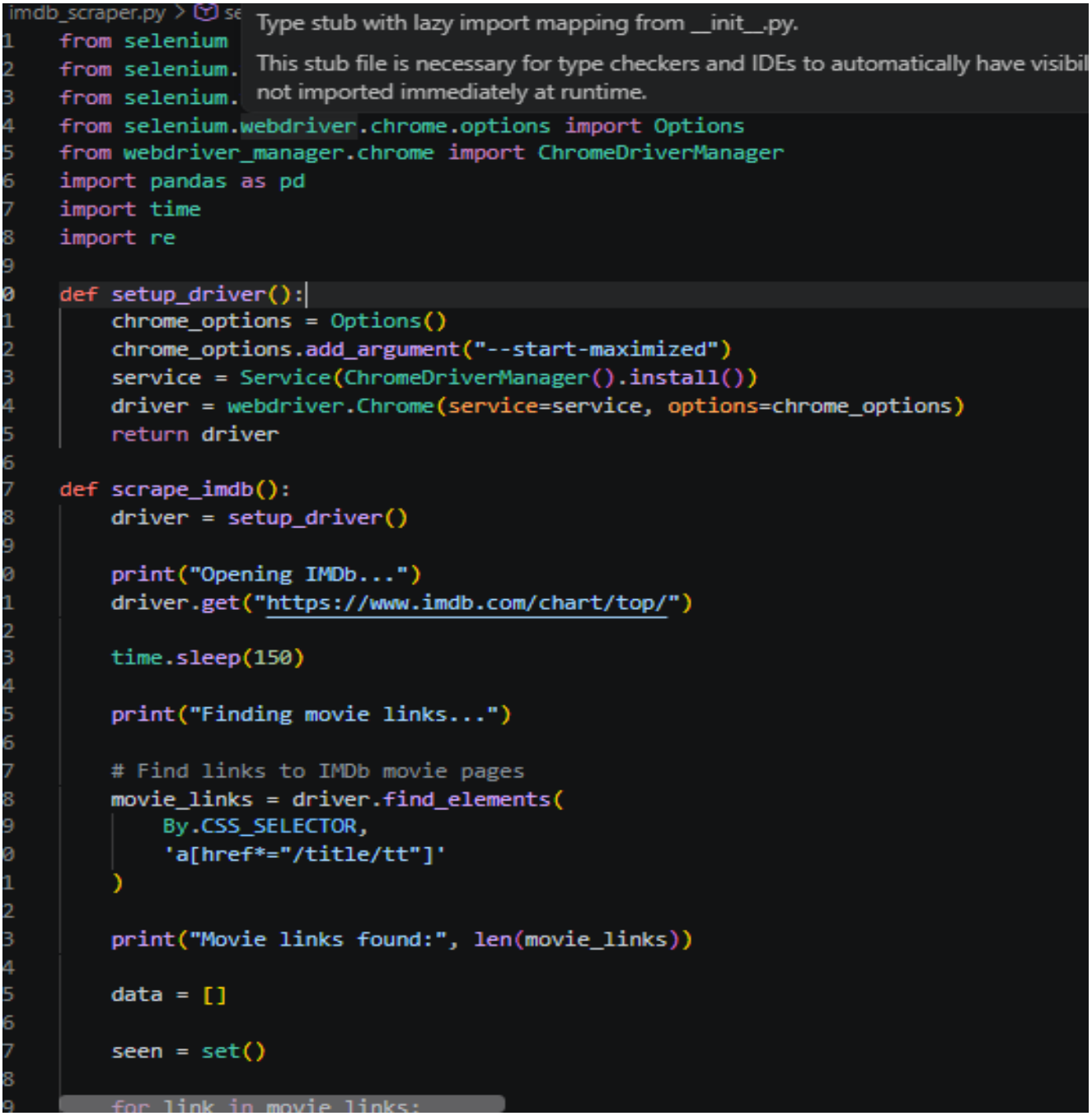
A	B	C	D
Rank	Title	Year	Rating
1	The Shawshank Redemption	1994	9.3
2	The Godfather	1972	9.2
3	The Dark Knight	2008	9.1
4	The Godfather Part II	1974	9.0
5	The Lord of the Rings: The Return of the King	2003	9.0
6	12 Angry Men	1957	9.0
7	Schindler's List	1993	9.0
8	The Lord of the Rings: The Fellowship of the Ring	2001	8.9
9	Pulp Fiction	1994	8.8
10	The Lord of the Rings: The Two Towers	2002	8.8
11	The Good, the Bad and the Ugly	1966	8.8
12	Forrest Gump	1994	8.8
13	Fight Club	1999	8.8
14	Inception	2010	8.8
15	Star Wars: Episode V - The Empire Strikes Back	1980	8.7

9.CONCLUSION

- The IMDb Top 250 Movie Rating Website successfully collects, processes, and displays movie information using IMDb datasets. It provides an easy-to-use platform for viewing, searching, and sorting top-rated movies, demonstrating the practical use of Python, Pandas, Flask, and web technologies.

10.CODE

- Uses Selenium for browser automation, Pandas for data processing, and supporting modules for delays, element selection, driver setup, and pattern matching.



Scraping Part (main part):

- The function opens IMDb, extracts unique movie links, and collects each movie's title, release year, and rating using Selenium and regular expressions.

  It stores up to 250 movie records, closes the browser, and returns the collected data.

```python
def scrape_imdb():
    for link in movie_links:

        try:
            title = link.text.strip()
            href = link.get_attribute("href")

            # Ignore empty links
            if not title:
                continue

            # Avoid duplicate movie links
            if href in seen:
                continue

            seen.add(href)

            # Get parent/container text
            parent = link.find_element(
                By.XPATH,
                "./ancestor::li[1]"
            )

            text = parent.text.strip()

            print("\nMovie:")
            print(text)

            # Find year directly from IMDB data
            year_match = re.search(r'(?<!\d)((?:19|20)\d{2})', text)
            if year_match:
                year = year_match.group()
            else:
                year = "N/A"

            # Find rating
            rating_match = re.search(
                r"\b([0-9]\.[0-9])\b",
```

11.OUTPUT

- Automated Data Extraction – Successfully extracts IMDb movie details using Selenium.
- Reliable Movie Data – Collects movie title, year, rating, and ranking.
- Structured Dataset – Stores the extracted data in CSV/Excel format.
- Easy Data Analysis – The collected dataset can be used for dashboards, analysis, and future applications.



- The web dashboard provides a user-friendly interface to display the IMDb Top 250 movies with their rank, title, release year, and rating. It allows users to easily search and sort movies, making movie information accessible and convenient to explore.